
Symmetries in Physics — Exercise Sheet 4

Schur's lemma and the group algebra

Variant: Questions only

Exercise 1 Use Schur's lemma to prove: (i) every irreducible complex representation of an abelian group is one-dimensional; (ii) in any irreducible representation, central elements act as scalars.

Exercise 2 Take the two-dimensional irreducible representation of $S_3 \cong D_3$ by rotations r^k through $2\pi k/3$ and reflections sr^k , with $s = \text{diag}(1, -1)$ as on Sheet 1. Writing $D_{\mu\nu}(g)$ for its matrix elements and $n = 2$ for its dimension, verify the orthogonality of matrix elements in two instances: $\sum_g |D_{11}(g)|^2 = |G|/n$ and $\sum_g D_{11}(g)\overline{D_{12}(g)} = 0$.

Exercise 3 Granting the lecture's isomorphism $\mathbb{C}[G] \cong \bigoplus_i M_{d_i}(\mathbb{C})$, verify the dimension count $|G| = \sum_i d_i^2$ for S_3 and for D_4 , and deduce how many times each irreducible occurs in the regular representation.

Exercise 4 On $\mathbb{Z}/n\mathbb{Z}$ define the convolution $(f * h)(x) = \sum_y f(y)h(x - y)$ and the Fourier transform $\widehat{f}(k) = \sum_x f(x)\omega^{-kx}$, $\omega = e^{2\pi i/n}$. Prove the convolution theorem $\widehat{f * h} = \widehat{f}\widehat{h}$, and interpret it as the statement that the basis of characters χ_k (the vectors v_k of Sheet 3, Exercise 1 — the *Fourier basis*) simultaneously block-diagonalises the regular representation.

Exercise 5 The quaternion group is $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$. (i) Find all one-dimensional representations. (ii) Deduce from $\sum d_i^2 = 8$ that exactly one two-dimensional irreducible remains, and realise it with Pauli matrices. (iii) Verify its irreducibility by a commutant argument: show that only scalars commute with all the representing matrices, and explain (using Maschke's theorem) why a reducible representation would admit a non-scalar commuting operator.

Exercise 6 Verify, for D_4 , the lecture's counting theorem: the number of irreducible representations equals the number of conjugacy classes.